INFORMATION RETRIEVAL – LAB PRACTICALS

Algorithms and flowcharts for all five practicals | Simple exam-revision format

Note: These are simplified study algorithms and flowcharts based on the five practical topics in your IR Lab Manual. For Practical 3, the manual specifies the Bayesian-network task and Heart Disease dataset, but does not prescribe one fixed coding implementation; the steps show the general model-building and inference process.

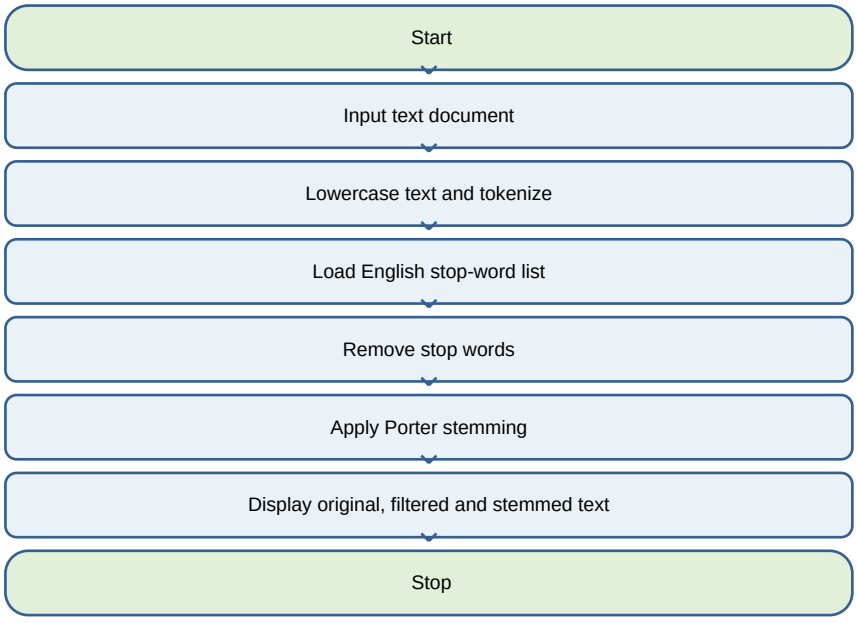
Practical 1 – Text Preprocessing: Stop-word Removal and Stemming

Problem Statement: Preprocess a text document by removing stop words and applying stemming.

Algorithm

1. Start; take input text document.
2. Convert text to lowercase and tokenize it into words.
3. Load the English stop-word list.
4. Compare each token with the stop-word list and remove stop words.
5. Apply Porter Stemmer to each remaining word.
6. Display original text, filtered words and stemmed words.
7. Stop.

Flowchart



Quick revision: 1 = preprocessing; 2 = word-to-document index; 3 = probability-based diagnosis; 4 = bottom-up clustering; 5 = webpage ranking.

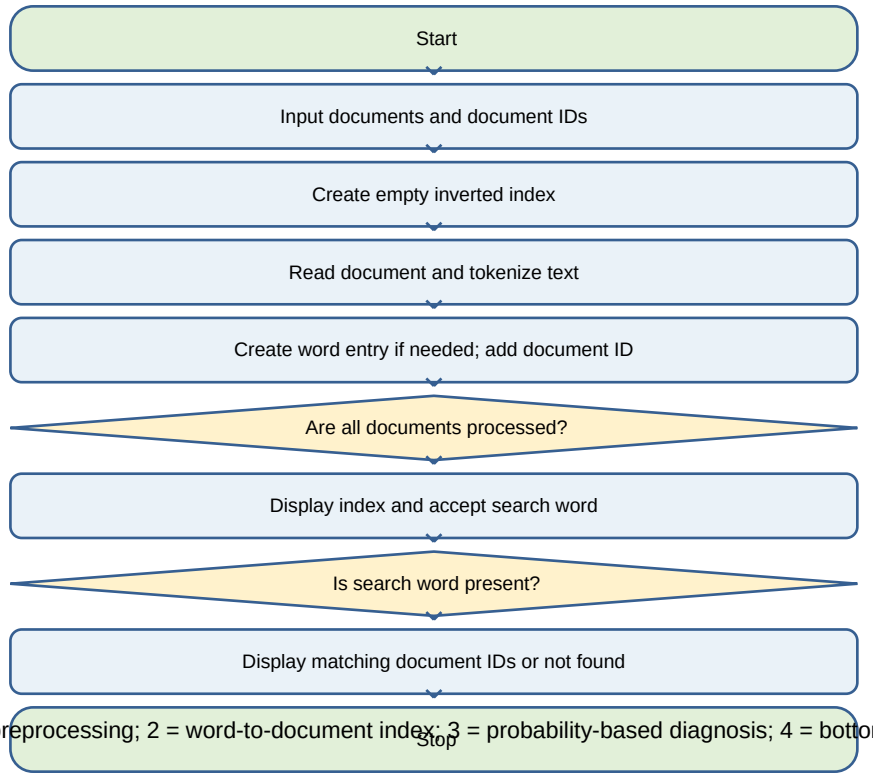
Practical 2 – Inverted Index (Inverted Files)

Problem Statement: Implement an inverted index to retrieve documents containing a given search term.

Algorithm

1. Start and input documents with their document IDs.
2. Create an empty inverted-index dictionary.
3. For each document, convert text to lowercase and tokenize it.
4. For each word, create an index entry if it does not exist.
5. Add the current document ID to that word's posting list (avoid duplicate IDs).
6. Display the inverted index.
7. Accept a search word; display matching document IDs if found, otherwise display "Word not found".
8. Stop.

Flowchart



Quick revision: 1 = preprocessing; 2 = word-to-document index; 3 = probability-based diagnosis; 4 = bottom-up clustering; 5 = webpage ranking.

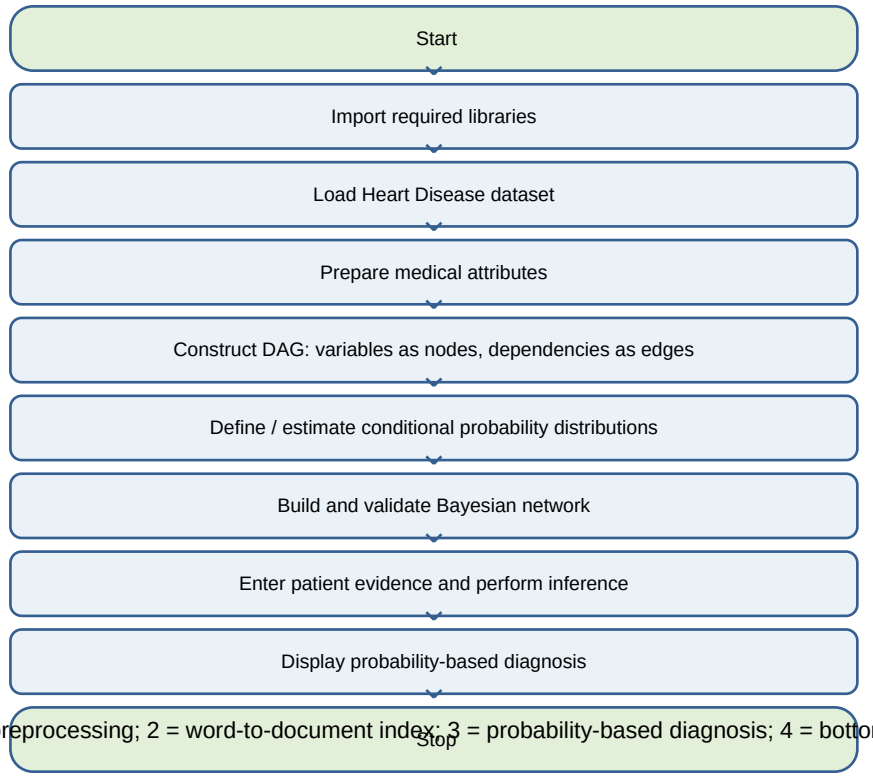
Practical 3 – Bayesian Network for Heart Disease Diagnosis

Problem Statement: Construct a Bayesian network using medical data and demonstrate diagnosis using the standard Heart Disease dataset.

Algorithm

1. Start and import required libraries.
2. Load the Cleveland Heart Disease dataset and inspect its medical attributes.
3. Prepare the data required by the Bayesian-network model.
4. Construct a directed acyclic graph (DAG): nodes represent variables and edges represent conditional dependencies.
5. Define or estimate conditional probability distributions (CPDs).
6. Build and validate the Bayesian network.
7. Provide patient evidence and perform probabilistic inference.
8. Display the probability-based diagnosis/result and stop.

Flowchart



Quick revision: 1 = preprocessing; 2 = word-to-document index; 3 = probability-based diagnosis; 4 = bottom-up clustering; 5 = webpage ranking.

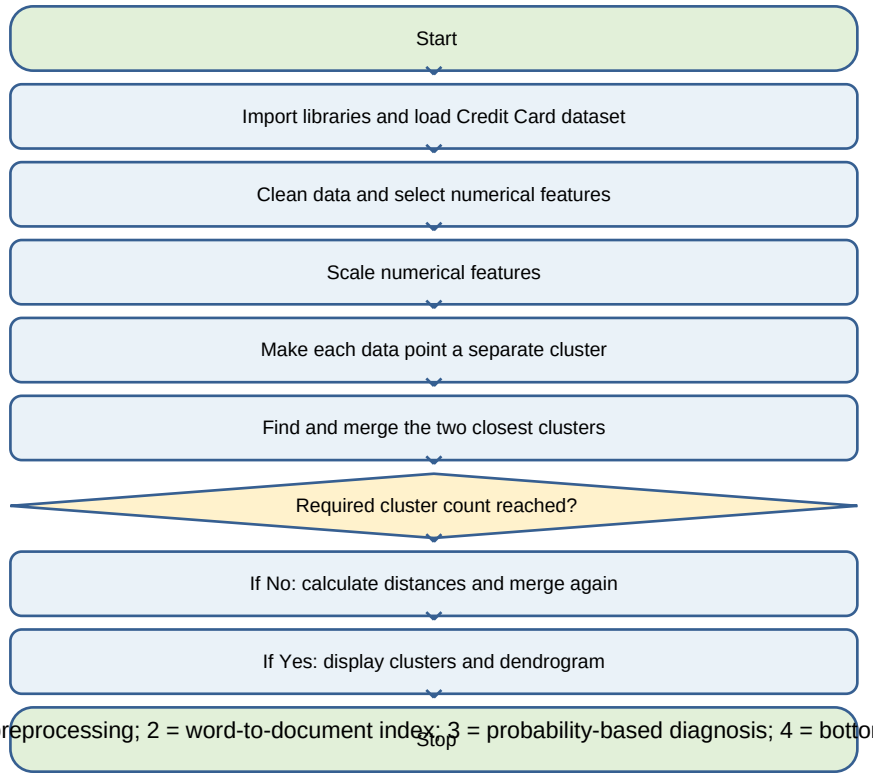
Practical 4 – Agglomerative Hierarchical Clustering

Problem Statement: Implement Agglomerative Hierarchical Clustering using an appropriate dataset (Credit Card Dataset in the manual).

Algorithm

1. Start and import required libraries.
2. Load the Credit Card dataset.
3. Prepare data: remove identifier columns and handle missing values where required.
4. Scale numerical features before distance calculation.
5. Initially treat every data point as a separate cluster.
6. Calculate distances and merge the two closest clusters.
7. Repeat until the required number of clusters (or one cluster) is reached.
8. Display cluster assignments and dendrogram; stop.

Flowchart



Quick revision: 1 = preprocessing; 2 = word-to-document index; 3 = probability-based diagnosis; 4 = bottom-up clustering; 5 = webpage ranking.

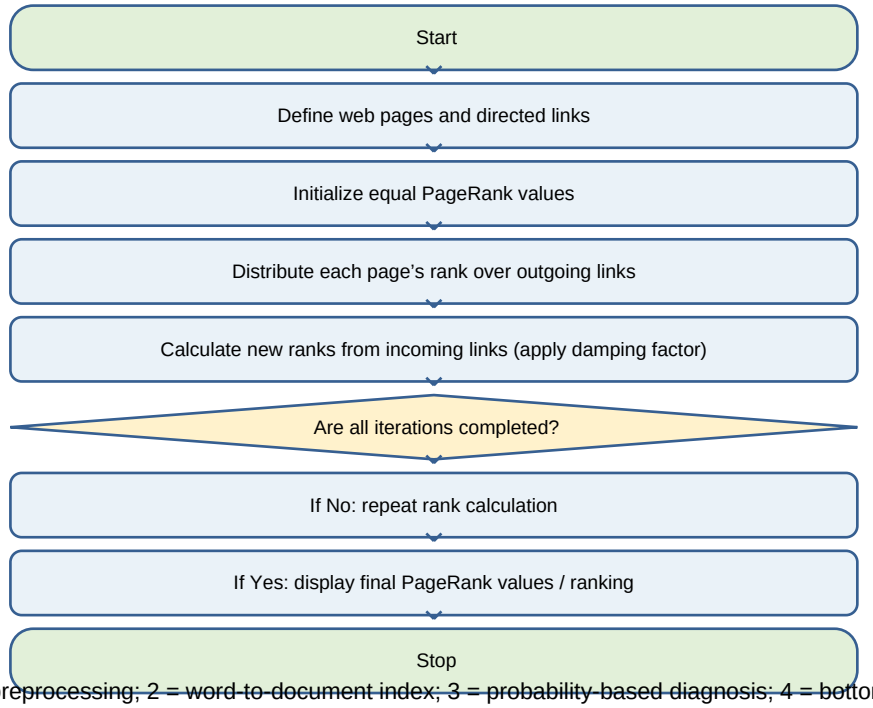
Practical 5 – PageRank Algorithm

Problem Statement: Implement PageRank to calculate the importance of web pages from their incoming and outgoing links.

Algorithm

1. Start and define web pages and directed links.
2. Let N be the total number of pages; initialize every page with equal rank (four pages: 0.25 each).
3. Distribute each page's current rank equally among its outgoing links.
4. For each page, add contributions received from all pages linking to it; apply damping factor in the full formula.
5. Repeat rank calculation for the specified number of iterations.
6. Display final PageRank values and compare pages by rank.
7. Stop.

Flowchart



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